Part IA Nat Sci and Comp Sci

Mathematical Methods II (Course B), Lent term 2026

Example sheet 1 : Ordinary Differential Equations

If you find errors on this sheet then please inform Robert Jack on `rlj22@cam.ac.uk`

Skills section

Questions in this section are intended to give practice in routine calculations. They start with revision exercises on integration, because this is an important aspect of solving differential equations.

S1. Find the indefinite integral $\displaystyle\int f(x)\,\mathrm{d}x$ for the following choices of $f = f(x)$:

(a) $(1+x)^{1/4}$ (b) $\dfrac{2+x}{(1+x)^2}$

(c) $2x(3+x^2)$ (d) $2x\sin(x^2)$

(e) $\cot x$ (f) $x^2 \sin x$ [*Hint: Integration by parts*]

(g) $\sin^3 x$ [*Hint: Trig. identity*] (h) $\dfrac{x}{(1-x)(2-x)}$ [*Hint: Partial fractions*]

(i) $\dfrac{1}{1+x^2}$ [*Hint: Substitution*] (j) $\sin\sqrt{1-x}$ [*Hint: Substitution*]

S2. Verify the given solutions for the following differential equations. In each case $c, d, \ldots$ are arbitrary constants.

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x\,,$ $y = \frac{1}{2}x^2 + c\,.$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = y\,,$ $y = c\,\mathrm{e}^x\,.$

(c) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 5\dfrac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0\,,$ $y = c\,\mathrm{e}^{-4x} + d\,\mathrm{e}^{-x}\,.$

(d) $x^2\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 6x\dfrac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0\,,$ $y = cx^{-4} + dx^{-1}\,.$

(e) $\dfrac{\mathrm{d}^4y}{\mathrm{d}x^4} - y = 0\,,$ $y = c\sin x + d\cos x + f\,\mathrm{e}^x + g\,\mathrm{e}^{-x}\,.$

(f) $\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 4y^3 = \dfrac{8}{x^6}\,,$ $y = x^{-2}\,.$

S3. Solve by separation of variables:

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{x^3}{(y+1)^2}$.

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{4y}{x(y-3)}$.

S4. Use integrating factors to solve:

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 4x$.

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + (2-3x^2)x^{-3}y = 1$.

Standard questions

Questions in this Section are intended to illustrate the key ideas and principles that were presented in lectures, in order to solve a variety of different ODEs. Unless otherwise stated, you should find the general solutions.

5. Solve by change of variables and separation:

$$(x+y+1)^2\,\frac{\mathrm{d}y}{\mathrm{d}x} + (x+y+1)^2 + x^3 = 0\,.$$

[Hint: let $u = x+y+1$.]

6. Solve by change of variables and the use of integrating factors:

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} - y = xy^5$.

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = y^2(\cos x - \sin x)$.

7. Consider the ODE

$$(y-x)\frac{\mathrm{d}y}{\mathrm{d}x} + (2x+3y) = 0\,.$$

Write this equation in the form $\frac{\mathrm{d}y}{\mathrm{d}x} = F(y/x)$ and hence derive:

$$y^2 + 2xy + 2x^2 = A\exp\left[4\arctan\left(\frac{y}{x}+1\right)\right]$$

where A is an arbitrary constant.

8. Solve the following ODEs (substitutions will be useful):

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y}{x} + \tan\left(\dfrac{y}{x}\right)$.

(b) $(\ln y - x)\dfrac{\mathrm{d}y}{\mathrm{d}x} - y\ln y = 0$.

(c) $xy\dfrac{\mathrm{d}y}{\mathrm{d}x} + (x^2+y^2+x) = 0$.

9. Find particular solutions to the following equations, given initial conditions $y(0) = 0$, $y'(0) = 1$.

(a) $\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 5\dfrac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0\,.$

(b) $\left(\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} + n^2\right) y = 0\,.$

(c) $\left(\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} + 2\dfrac{\mathrm{d}}{\mathrm{d}x} + 4\right) y = 0\,.$

(d) $\left(\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} + 9\right) y = 18\,.$

(e) $\left(\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} - 3\dfrac{\mathrm{d}}{\mathrm{d}x} + 2\right) y = \mathrm{e}^{5x}\,.$

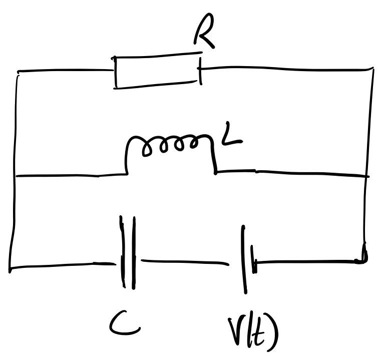
Also find the general solutions to

(f) $\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 2\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = 0\,.$

(g) $\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 2\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = \mathrm{e}^{2x}\,.$

[note for supervisors: part (g) has been modified with respect to the 2025 edition, see also Q11.]

10. The electrical circuit pictured below allows exploration of transient and resonance phenomena. There is a resistor with resistance R, an inductor with inductance L, and a capacitor with capacitance C. There is a power supply which sets a time dependent voltage $V(t)$.



The charge on the capacitor is $q = q(t)$, it obeys the ordinary differential equation

$$L\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + \frac{L}{RC}\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{C} + V + \frac{L}{R}\frac{\mathrm{d}V}{\mathrm{d}t} = 0\,.$$

were L, C, R are positive constants.

Consider the situation where the voltage starts at a negative initial value, and is suddenly changed to zero at time $t = 0$:

$$V(t) = \begin{cases} -V_0 & t < 0 \\ 0 & t \geq 0 \end{cases}$$

(a) Show that for $t < 0$ one can solve the ODE as $q(t) = Q$ (independent of t). How does Q depend on V_0?

(b) To find $q(t)$ for $t > 0$, we solve the ODE with $V(t) = 0$ [so also $(\mathrm{d}V/\mathrm{d}t) = 0$] and initial conditions $q(0) = Q$ and $\frac{\mathrm{d}q}{\mathrm{d}t}(0) = -Q/RC$. Find the solution in the case $L = 8R^2C$.

(c) Find the solutions for $L = 4R^2C$ and $L = 2R^2C$. Comment on the differences between the three solutions that you have found.

11. Consider the equation

$$y'' - (2 + c)y' + (1 + c)y = \mathrm{e}^{(1+2c)x}$$

and assume $c \neq 0$. Find the complementary function and a particular integral. Hence show that the general solution can be expressed as

$$y(x) = A\,\mathrm{e}^x + B\frac{\mathrm{e}^x}{c}(\mathrm{e}^{cx} - 1) + \frac{\mathrm{e}^x}{2c^2}(\mathrm{e}^{2cx} - 2\,\mathrm{e}^{cx} + 1)\,,$$

where A, B are arbitrary constants.

Evaluate this general solution in the limit $c \to 0$. Use this result to identify the particular integral and complementary function for the equation

$$y'' - 2y' + y = \mathrm{e}^x$$

Compare your answer to this question with Q9(f,g). What happens if you use the standard method [of trial solutions] to look for the particular integral in this case?

Find a particular integral for the equation

$$y'' - 3y' + 2y = \mathrm{e}^x\ .$$

(Hint: you should find that the trial solution $y = \mathrm{e}^x$ is not sufficient.)

12. Show that the equation

$$-\frac{\eta}{2}\frac{\mathrm{d}C}{\mathrm{d}\eta} = \frac{\mathrm{d}^2C}{\mathrm{d}\eta^2}$$

can be written as

$$\frac{\mathrm{d}}{\mathrm{d}\eta}\ln\left(\frac{\mathrm{d}C}{\mathrm{d}\eta}\right) = -\frac{\eta}{2}\,.$$

Hence show that if $\frac{\mathrm{d}C}{\mathrm{d}\eta}(0) = A$ and $C(0) = B$ then

$$C(\eta) = B + A\int_0^\eta \exp(-t^2/4)\,\mathrm{d}t\,.$$

13. (a) Solve the pair of differential equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = ax\,, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = ay + bx\,,$$

subject to the initial conditions $x(0) = 2$, $y(0) = 1$. (Here, a and b are constants.)

(b) The differential equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x - xy\,, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = -y + xy\,,$$

give the behaviour of x and y as a function of t. Use them to derive a single differential equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y)\,.$$

Solve this equation and use the result to explain that

$$g(x, y) = \frac{\mathrm{e}^{x+y}}{xy}$$

is independent of t.